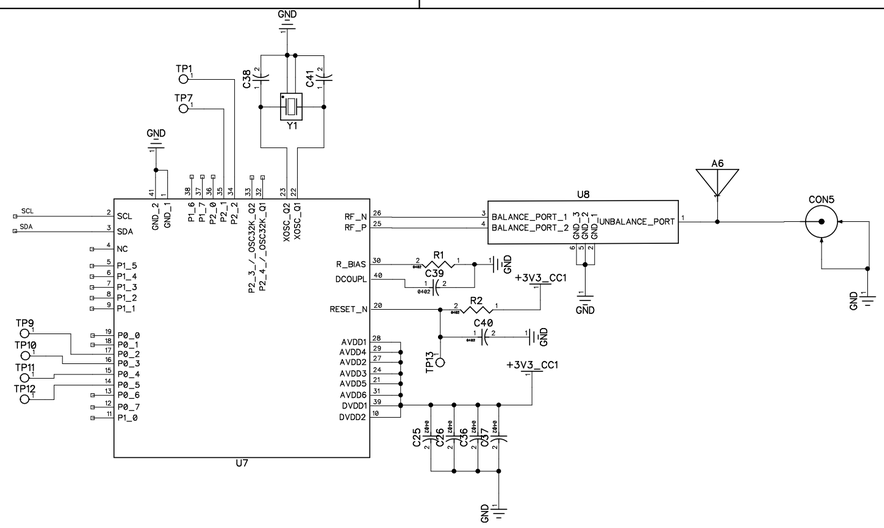
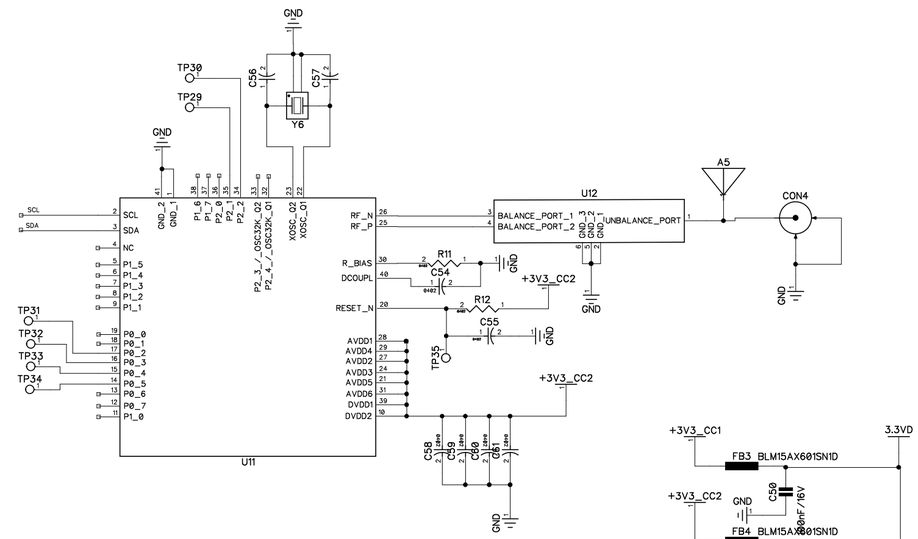
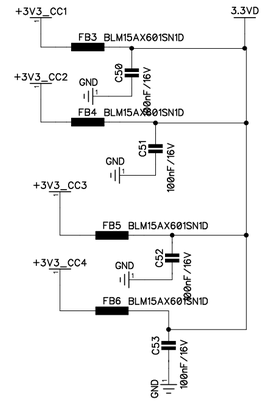
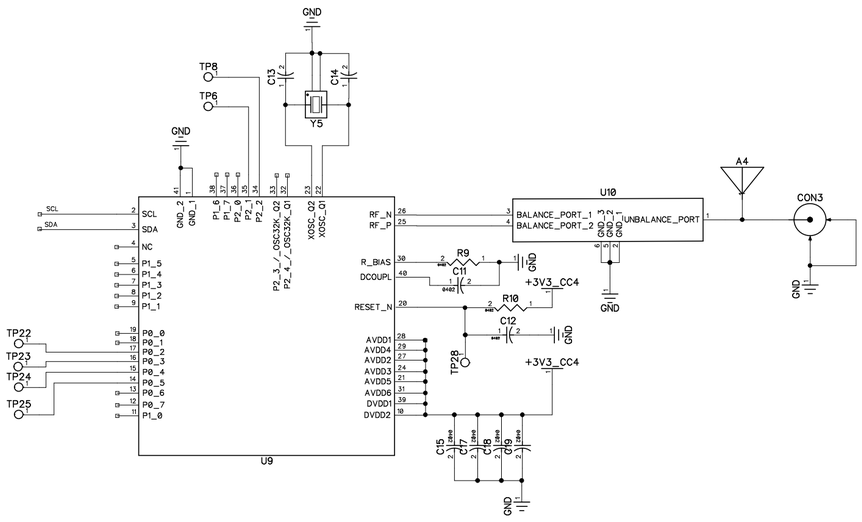
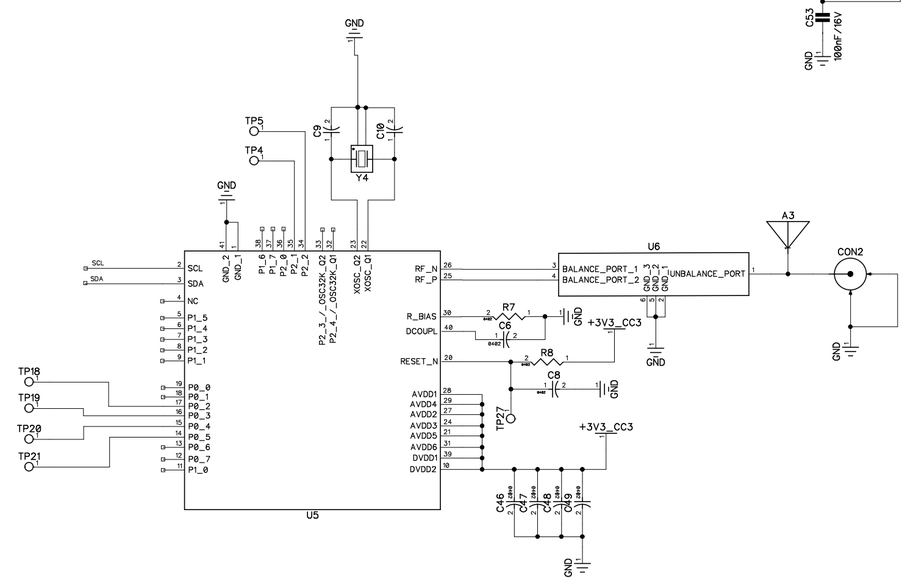
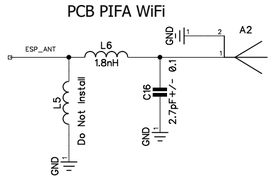

GND
TP1
TP7
C38
C41
Y1
GND
U7
SCL
SDA
NC
P1_5
P1_4
P1_3
P1_2
P1_1
P0_0
P0_1
P0_2
P0_3
P0_4
P0_5
P0_6
P0_7
P1_0
GND_2
GND_1
P1_6
P1_7
P2_0
P2_1
P2_2
P2_3_/_OSC32K_Q2
P2_4_/_OSC32K_Q1
XOSC_Q2
XOSC_Q1
TP9
TP10
TP11
TP12
RF_N
RF_P
R_BIAS
DCOUPL
RESET_N
AVDD1
AVDD4
AVDD2
AVDD3
AVDD5
AVDD6
DVDD1
DVDD2
R1
C39
R2
C40
TP13
+3V3_CC1
C25
C26
C16
C27
U8
BALANCE_PORT_1
BALANCE_PORT_2
UNBALANCE_PORT
A6
CON5

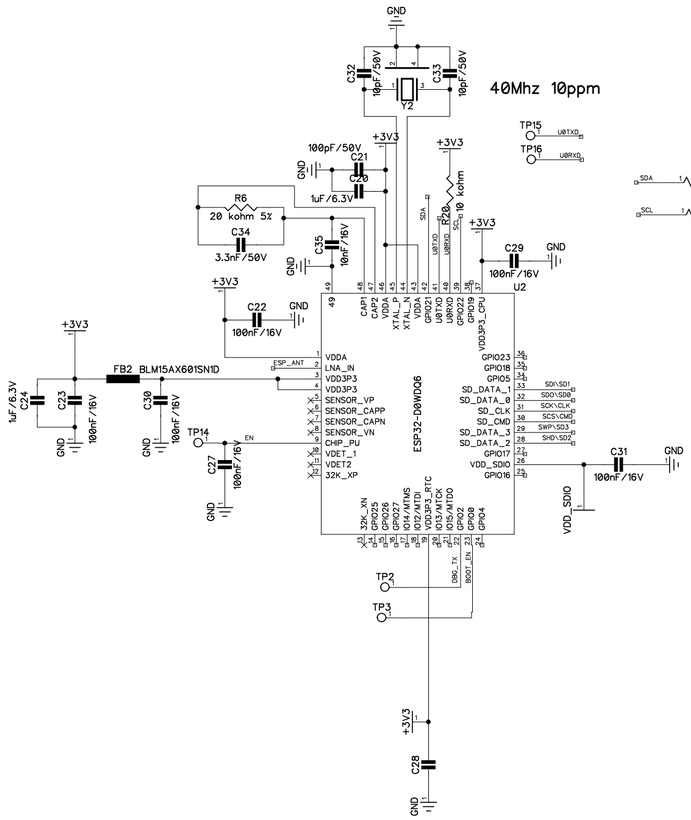

GND
TP30
TP29
C56
C57
Y6
U11
SCL
SDA
NC
P1_5
P1_4
P1_3
P1_2
P1_1
P0_0
P0_1
P0_2
P0_3
P0_4
P0_5
P0_6
P0_7
P1_0
GND_2
GND_1
P1_6
P1_7
P2_0
P2_1
P2_2
P2_3_/_OSC32K_Q2
P2_4_/_OSC32K_Q1
XOSC_Q2
XOSC_Q1
TP31
TP32
TP33
TP34
RF_N
RF_P
R_BIAS
DCOUPL
RESET_N
AVDD1
AVDD4
AVDD2
AVDD3
AVDD5
AVDD6
DVDD1
DVDD2
R11
C54
R12
C55
TP35
+3V3_CC2
C58
C59
C60
C61
U12
BALANCE_PORT_1
BALANCE_PORT_2
UNBALANCE_PORT
A5
CON4

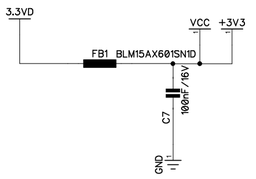

+3V3_CC1
3.3VD
FB3 BLM15AX601SN1D
+3V3_CC2
GND
C50
100nF/16V
FB4 BLM15AX601SN1D
C51
100nF/16V
+3V3_CC3
FB5 BLM15AX601SN1D
+3V3_CC4
C52
100nF/16V
FB6 BLM15AX601SN1D
C53
100nF/16V

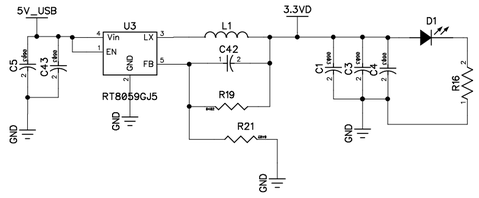

GND
TP8
TP6
C13
C14
Y5
U9
SCL
SDA
NC
P1_5
P1_4
P1_3
P1_2
P1_1
P0_0
P0_1
P0_2
P0_3
P0_4
P0_5
P0_6
P0_7
P1_0
GND_2
GND_1
P1_6
P1_7
P2_0
P2_1
P2_2
P2_3_/_OSC32K_Q2
P2_4_/_OSC32K_Q1
XOSC_Q2
XOSC_Q1
TP22
TP23
TP24
TP25
RF_N
RF_P
R_BIAS
DCOUPL
RESET_N
AVDD1
AVDD4
AVDD2
AVDD3
AVDD5
AVDD6
DVDD1
DVDD2
R9
C11
R10
C12
TP28
+3V3_CC4
C15
C17
C18
C19
U10
BALANCE_PORT_1
BALANCE_PORT_2
UNBALANCE_PORT
A4
CON3

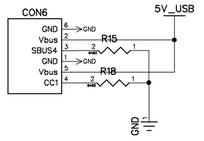

GND
TP5
TP4
C9
C10
Y4
U5
SCL
SDA
NC
P1_5
P1_4
P1_3
P1_2
P1_1
P0_0
P0_1
P0_2
P0_3
P0_4
P0_5
P0_6
P0_7
P1_0
GND_2
GND_1
P1_6
P1_7
P2_0
P2_1
P2_2
P2_3_/_OSC32K_Q2
P2_4_/_OSC32K_Q1
XOSC_Q2
XOSC_Q1
TP18
TP19
TP20
TP21
RF_N
RF_P
R_BIAS
DCOUPL
RESET_N
AVDD1
AVDD4
AVDD2
AVDD3
AVDD5
AVDD6
DVDD1
DVDD2
R7
C6
R8
C8
TP27
+3V3_CC3
C46
C47
C48
C49
U6
BALANCE_PORT_1
BALANCE_PORT_2
UNBALANCE_PORT
A3
CON2


PCB PIFA WiFi
ESP_ANT
L6
1.8nH
GND
A2
L5
Do Not Install
C16
2.7pF+/- 0.1
GND
GND


GND
C32
10pF/50V
C33
10pF/50V
Y2
40Mhz 10ppm
+3V3
100pF/50V
C21
GND
C20
1uF/6.3V
R6
20 kohm 5%
C34
3.3nF/50V
C35
10nF/16V
GND
+3V3
R20
10 kohm
SDA
SCL
U0TXD
U0RXD
+3V3
C29
100nF/16V
GND
U2
+3V3
C22
100nF/16V
GND
+3V3
1uF/6.3V
C24
C23
100nF/16V
GND
FB2 BLM15AX601SN1D
C30
100nF/16V
GND
TP14
EN
C27
100nF/16V
GND
ESP32-D0WDQ6
VDDA
LNA_IN
VDD3P3
VDD3P3
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
CHIP_PU
VDET_1
VDET_2
32K_XP
32K_XN
GPIO25
GPIO26
GPIO27
GPIO14/MTMS
IO12/MTDI
VDD3P3_RTC
IO13/MTCK
IO15/MTDO
GPIO2
GPIO8
GPIO4
CAP1
CAP2
VDDA
XTAL_P
XTAL_N
VDDA
GPIO21
U0TXD
U0RXD
GPIO22
GPIO19
VDD3P3_CPU
GPIO23
GPIO18
GPIO5
SD_DATA_1
SD_DATA_0
SD_CLK
SD_CMD
SD_DATA_3
SD_DATA_2
GPIO17
VDD_SDIO
GPIO16
SD1\SD1
SD0\SD0
SCK\CLK
SCS\CMD
SWP\SD3
SHD\SD2
C31
100nF/16V
GND
VDD_SDIO
TP2
DBG_TX
TP3
BOOT_EN
+3V3
C28
GND
TP15
U0TXD
TP16
U0RXD
SDA
R3
SCL
R5
+3V3
3V3
R14 10 kohm
BOOT_EN
R13 10 kohm
DBG_TX
R4 10 kohm
EN
Flash Memory
VDD_SDIO
U4
SCS\CMD
SDO\SD0
SWP\SD3
GND
SCS/CMD
SDO/SD0
SWP/SD3
GND
Vcc
SD2/SHD
CLK
SD1/SDI
SHD\SD2
SCK\CLK
SD1\SDI
W25Q128JVSIQ
GND


3.3VD
VCC
+3V3
FB1 BLM15AX601SN1D
C7
100nF/16V
GND


5V_USB
C5
C43
GND
U3
Vin
EN
LX
GND
FB
RT8059GJ5
GND
L1
3.3VD
C42
R19
R21
GND
D1
C1
C2
C3
C4
R16
GND


CON6
GND
Vbus
SBUS4
GND
Vbus
CC1
GND
GND
5V_USB
R15
R18
GND